

Architectural Particular Specification

FIRESTOPPING

SECTION 6.4

Rev	Date	Description
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FIRESTOPPING

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The work shall include supply and installation of fire rated enclosure, including but not limited to fire resistance boards, associated support frames, access hatches and fire-rated sealants, for the following areas:-
1. protected lift lobbies / corridor
 2. escape staircase
 3. services passing through another plant rooms
 4. For maintenance of compartmentation fire enclosure
- B. The Work of this Section includes, but is not limited to, the furnishing and installing of the following :
1. Continuous firestopping and required accessories in all fire-rated assemblies matching the required fire-resistance requirement.
 2. Fire rated enclosures to ductwork and other services as indicated on the consulting engineers' documents.
 3. Continuous fire-rated barrier seals against fire, smoke, toxic fumes and hose stream water.
 4. Firestopping accessories including restricting collars, composite intumescent cover sheet and endothermic mat, mineral wool packing, moldable putty, fasteners with washers, and other ancillary items.
 5. Other firestopping and accessories indicated on the Contract Documents.
 6. Proprietary pipe, cable tray, flue, duct, conduit, telephone, cable fire/smoke penetrations barriers through fire rated- assemblies.
 7. Intumescent fire seals, mastics, smoke control seals, fire rated-resistant glazing gaskets and channels for installation on fire-rated doors in as stipulated elsewhere under Contract.

1.02 REFERENCES

- A. British Standards:
1. BS 476: Part 4: 1970 (1984)-Non-Combustibility Test For Materials.
 2. BS 476: Part 6: 1989-Method of Test for Fire Propagation for products.
 3. BS 476: Part 7: 1987-Method of Classification of the Surface Spread of Flame of Products.
 4. BS 476: Part 20: 1987-Method of Determination of the Fire Resistance of Elements of Construction.
 5. BS 476: Part 23:1987 - Method of Determination of the Contribution of Components to the Fire Resistance of a Structure.
 6. BS 576 Part 24: 1987 Method for Determination of the Fire Resistance of Ventilation Ducts.
 7. BS 476: Part 31.1: 1983-Method of Measuring Smoke Penetration Through Doorsets and Shutter Assemblies.

B. American Standards :

1. ASTM E119-88: Fire Tests of Building Construction and Materials.
2. ASTM E814-94b: Fire Tests of Through Penetration Firestops (Hose Stream Test).
3. UL 1479 - Firestops, Fire Tests of Through-Penetrations.
4. UL Fire Resistance Directory of Through - Penetration Firestop Systems, latest edition.

1.03 QUALITY ASSURANCE

- A. Qualification of firestop specialist sub-contractor: Company specializing in firestopping and fire proofing application with minimum of five (5) years documented experience, certified and approved by the Firestop Manufacturers specified herein.
- B. Qualification of firestop manufacturers:
1. Manufacturers capable of providing field service representation during Contract Documents review, construction, approving and training acceptable installer, application method and conducting final inspection of firestopping system/assembly.
 2. The Firestop Manufacturer shall have approved Testing Agency to test fire-resistance of proposed firestop and smoke seal design and mock-up test examples.
- C. Regulatory requirements:
1. All materials provided under this section to be approved by the statutory authorities and bear the certified label as required.
 2. All firestop and barrier seal system materials shall be certified by the British Board of Agrément (BBA), tested to BS 476: Part 20 and Part 22, and all referenced Standards cited herein. Alternatively, all firestop and barrier seal system materials shall have been tested, approved and listed by U.S. Underwriter's Laboratory (UL) in UL Fire Resistance Directory of Through-Penetration Firestop Systems for applications indicated, with system or material capable of passing ASTM E-814 (ANSI/UL 1479) Standard Method of Fire Tests for Through Penetration Firestops up to the specified fire resistance.
- D. During the Progress of the Project Construction, the Contractor shall ensure that all penetrations, gaps, cavities, voids, enclosed spaces, etc. at every floor level and locations are firestopped and barrier sealed before they are enclosed or made inaccessible by subsequent Contract Work.

1.04 PERFORMANCE REQUIREMENTS

A. Fire Resistance Period

All fire rated assembly or enclosure shall meet all the relevant requirement of the Fire Service Department and the Building Authority of Hong Kong and shall attain minimum fire resistance rating (FRR) as shown below

AREA	F.R.R.
Enclosure around air ducts, pipes or wires within Fire Escape Stair Enclosures, Smoke Lobbies, thru compartments enclosure, and Protected Corridors	-/120/120

- B. Carry out all necessary test and / or obtain all necessary certificates as required by the FSD and the Building Authority.
- C. All the fire rated assembly or enclosure should have FRR certificate complying with Buildings Department & Fire Services Department requirement, and BS 476: Part 20 to 24 or equivalent.

1.05 ENVIRONMENTAL REQUIREMENTS

- A. It is to be noted that the ceiling void and fan rooms on the project are used as return air plenum. All materials used in these locations are to be certified that insulation fibres will not be released into the air stream over the life time of the building.
- B. The following shall not be installed on the project, materials which are generally comprised of mineral fibres, either man-made or naturally occurring, which have a diameter of 3 microns or less and a length of 200 microns or less or which contain any fibres not sealed, encapsulated or otherwise stabilised to ensure that fibre migration is prevented.
- C. Do not install solvent curing firestop compounds and sealants in enclosed building spaces without adequate ventilation; conform with Manufacturer's recommendation.
- D. Maintain temperature and humidity recommended by Firestop and Barrier Sealant Manufacturers during and after installation, and while in storage.
- E. The special precautions recommended by the Manufacturer shall be strictly followed where flammable and hazardous materials are involved or released. Hazardous waste materials shall be legally transported and disposed of in strict conformance to local regulations.

1.06 SUBMITTALS

- A. Manufacturer's catalogues, test certificate, and installation method, material samples and shop drawings for all relevant materials and equipment shall be submitted to the Architect for review or approval. The extent of information or samples required shall be adequate and sufficient to demonstrate that the proposed system and materials are in compliance with the contract requirements. However, the Architect can request the contractor to submit any additional or supplementary information to substantiate the performance of the proposed system or material.
- B. No bulk materials and systems ordering work shall proceed prior to the Architect's confirmation of no adverse comment on the submitted information or samples. It is the contractor's sole responsibility to ensure that the required materials or systems can be ordered according to the contractor's master programme. No claim for any additional time or cost due to any delayed submission of a satisfactory system will be entertained.
- C. The following shall be submitted to the Architects for endorsement before mass ordering of materials and installation.
 - 1. Sample boards of each type of the fire-resistance board, supporting frame, access hatch fastener and sealant;
 - 2. Fire resistance certificates mentioned above;
 - 3. Shop drawings showing the arrangement and extent of all fire rated assembly and enclosure including all access panel locations.

D. Fire Stops to comply the following:-

1. Submit product and technical performance data, method statement for application, and samples for each type of firestop and barrier seal.
2. The Contractor shall submit required sets of firestop and barrier seal shop drawings with 1:20 scale elevation and plan drawings, and 1:2 scale details and drawings indicating locations of each type of firestop and barrier seal materials. Furnish shop drawings for all Flamebar BW11 fire protection installation and material specified herein and indicated on the Drawings.
3. Firestop and barrier seal testing submittals as required in Clause 1.06 of this Section.
4. Submit CERTIFIRE, UL, BBA and other approval certificates as required in this Section.
5. Samples of each type of Flamebar BW11, fixings, firestop and barrier seal intended to be used on the Project.

1.07 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.08 DELIVERY, STORAGE AND HANDLING

- A. Materials shall be delivered in Firestop and Barrier Seal Material Manufacturers' containers, or package, dry, undamaged, unopened, all clearly labelled with the Manufacturer's name product identification, expiration date, pot/shelf-life, and lot numbers intact. All UL, CERTIFIRE or BBA approval Certificate Labels shall be on all firestop and seal material. Any material without fire test certified label by a recognized Material Testing Laboratory shall be rejected and replaced.
- B. Materials shall be stored in strict accordance with the Manufacturer's printed instructions, copies of which shall be furnished to the Architect.
- C. Outdated materials may not be used. The Firestop and Barrier Seal Manufacturer's Technical Field Representative shall verify that all lots and/or shipments are from an authorized distributor and the firestop, barrier seal, and accessories are genuine products of the specified Firestop and Barrier Seal Manufacturers.
- D. Firestops shall comply the following:-
 1. Materials shall be delivered in Firestop and Barrier Seal Material Manufacturers' containers, or package, dry, undamaged, unopened, all clearly labelled with the Manufacturer's name product identification, expiration date, pot/shelf-life, and lot numbers intact. All UL, CERTIFIRE or BBA approval Certificate Labels shall be on all firestop and seal material. Any material without fire test certified label by an recognized Material Testing Laboratory shall be rejected and replaced.
 2. Materials shall be stored in strict accordance with the Manufacturer's printed instructions, copies of which shall be furnished to the Architect.
 3. Outdated materials may not be used. The Firestop and Barrier Seal Manufacturer's Technical Field Representative shall verify that all lots and/or shipments are from an authorized distributor and the firestop, barrier seal, and accessories are genuine products of the specified Firestop and Barrier Seal Manufacturers.

1.09 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.10 TESTING

(NOT USED)

1.11 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. Provide Manufacturer's ten (10) years warranty for all firestop and barrier seal materials against material defects, firestop and barrier performance, and workmanship.
- C. Nothing shall be done by the Contractor which will void any Manufacturer's Warranty, or invalidate the Project Building's Loss-Prevention Insurance coverage.

1.12 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS

- A. Subject to compliance with the requirements of the Project Contract Documents, provide firestop and barrier seal materials, and products by one, or more, of the following Manufacturers or equivalent approved by the Architect:

1. **[FS01], [FS02], [FS03], [FS04], [FS05], [FS06], [FS07], [FS08], [FS09]**
Hilti (Hong Kong) Ltd.
701-704 & 708B
Tower A, Manulife Financial Centre,
223 Wai Yip Street,
Kwun Tong
Hong Kong
Tel: (852) 8228 8118 Fax: (852) 2764 3234

2.02 PRODUCTS

- A. Fire-rated Enclosure for air ducts, pipes and wires

1. The contractor shall coordinate with the MVAC, Electrical and other relevant sub-contractors to determine the exact locations where fire rated enclosures are required. Shop drawings showing the extent of the fire rated enclosure shall be submitted to the Architect and the consultants for approval before material ordering and installation.
2. The contractor shall coordinate with the MVAC, electrical and other relevant sub-contractors such that the installation of the fire-rated assembly or enclosure will not hinder their relevant construction works or damage building service works.
3. The contractor shall provide access hatches on the fire-rated enclosure and at locations as deemed appropriate by the Architect and / or other consultant. Access hatch shall be of the same system as the rest of the fire-rated enclosure and by no means undermining the integrity of the overall fire enclosure system.
4. The fire-rated enclosure shall separate air ducts, pipes (except F.S. pipes) and all electrical works from the fire escape stair enclosure completely.

- B. Product of Fire Stops shall comply the following performance requirements:-

1. Provide and install firestop and barrier seal materials as required by code, and as shown on the Drawings, with ancillary sealant materials necessary for a proper, complete and continuous firestop and barrier seal against fire, smoke, toxic fumes and water from one or more of the approved Manufacturers listed in Clause 2.01 above.
2. Refer to Section 5.1 Carpentry, Laminate and Joinery for fire-retardant treatment of wood and board products.
3. Refer to Sections 7.1 and 7.2 for door and door frame fire stops and smoke seals.
4. Products for Firestopping:-
 - a. **[FS01]** Hilti CP 601S elastic firestop sealant
 - Silicon based elastic firestop sealant with movement capability $\pm 25\%$
 - Halogen and solvent free
 - Weather and UV resistant as per ASTM C920
 - Hardness; shore A index 30

- Maximum supporting gap width 100mm
 - F.R.R. up to -/240/-
 - Material class as per DIN 4102, P.1: B1
- b. **[FS02]** Hilti CP 606 flexible firestop sealant
- Acrylic based flexible firestop sealant with movement capability $\pm 10\%$
 - Halogen and solvent free
 - Maximum support gap width 30mm
 - F.R.R. up to -/240/-
 - Material class as per DIN 4102, P.1: B1
- c. **[FS03]** Hilti CP 648 intumescent pipe wraps
- Flexible intumescent graphite impregnated band
 - Chemical basis acrylic polymer
 - Halogen and plasticizer free
 - F.R.R. up to -/240/-
 - Material class as per DIN 4102, P.1; B2
- d. **[FS04]** Hilti CP 643 / CP 644 firestop jacket/collar
- Galvanized steel/A2 stainless steel housing and intumescent inserts
 - F.R.R. up to -/240/-
- e. **[FS05]** Hilti CP 636 firestop mortar
- High yield
 - Contains no asbestos, phenol or halogen constituents
 - Density of hardened mortar 820kg/m³
 - Mix ratio 3:1 (mortar to water)
 - F.R.R. up to -/240/-
 - Material class as per DIN 4102, P.1; A1
- f. **[FS06]** Firestop CP 617 intumescent acoustic putty pad
- Intumescent moldable putty pad for use with plastic or metal back-boxes
 - Chemical basis synthetic rubber
 - Electrically non-conductive
 - Smoke and gas tight
 - Acoustic data 64dB tested to DIN EN20140
 - F.R.R. up to -/120/-
- g. **[FS07]** Hilti CFS-BL intumescent firestop block
- Intumescent flexible brick composed of a two component polyurethane foam mixed with graphite
 - Halogen and solvent free
 - F.R.R. up to -/180/-

- h. **[FS08]** Hilti CP 670 fire safety coating
 - Odorless and solvent free
 - Chemical basis acrylic dispersion
 - F.R.R. up to -/240/-
- i. **[FS09]** Hilti CFS-SP WB firestop joint spray
 - Chemical basis acrylic water based dispersion with movement capability $\pm 50\%$
 - Fire tested according to BS EN 1364-4
 - Air tightness tested according to EN 12101-1
 - F.R.R. up to -/240/-
 - Movement capacity 50%
 - Paintable
- j. **[FS10]** Hilti CP611A intumescent firestop mastic
 - Graphite based intumescent firestop sealant with movement acceptance 10%
 - F.R.R. up to -/240/-
 - Material class as per DIN 4102, P.1:B2
 - Halogen and solvent free
 - Odorless
- k. **[FS11]** Graphite based intumescent firestop sealant
 - Graphite based intumescent firestop sealant with movement acceptance 10%
 - F.R.R. up to -/240/-
 - Material class as per DIN 4102, P.1:B2
 - Halogen and solvent free
 - Odorless

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify that surfaces, joint and penetration openings are ready to receive Work and field measurements are as shown on approved Shop Drawings and recommended by the Firestop & Barrier Seal Material Manufacturers.
- B. Beginning of installation means firestop specialist sub-contractor's acceptance of existing substrates and conditions.
 - 1. All gaps, cavities, voids, penetrations, enclosed spaces at every floor level and at all fire-rated floor/ceiling and wall assemblies shall be thoroughly inspected to ensure that firestop and barrier seals form continuous fire protection and separation, and prevent the purgation of fire, smoke and toxic fumes.
 - 2. Where openings in floors are in excess of 1.0m in any plane the fire sealing system shall be fitted with a support frame of minimum 50x50x3mm angle iron anchor bolted to the concrete slab.
 - 3. Sealing systems for floor openings in excess of 1.0m deep from the face of the penetrating services shall be load bearing and protected by steel plates or barriers.
 - 4. Fire sealing materials or systems shall be capable of being readily modified to accommodate future services.
 - 5. Cable trunking shall be sealed internally with a dry intumescent material being compressed between the cabling and the trunking cover.

3.02 PREPARATION

- A. Remove loose dirt, oil, and other contaminants from penetration surfaces.
- B. Place hangers or damming materials in penetration as required to hold firestopping materials. Mechanically attach firestop and barrier seal materials by approved methods to the surrounds to prevent dislodgment due to any air movement, intumescent activation, seismic and structural movement and deflections.

3.03 INSTALLATION

- A. General
 - 1. Strictly follow manufacturers' charts for appropriate material to achieve the required fire rating in various location and applications on the Project.
 - 2. Install fire resistance duct firestopping at all penetrations of fire rated wall and floor/ceiling assemblies in strict conformance with manufacturer's published instructions.
 - 3. Install firestop and barrier seal in strict conformance to UL Fire Resistance Directory for Through-Penetration Firestop Systems.
 - 4. Install fire resistant joint sealants per BS 6093: 1993 and Section 079000 requirements.
 - 5. Install fire protection assemblies and items per approved shop drawings.
 - 6. All installation shall comply to COP for Fire Safety in Building 2011.
 - 7. Provide all necessary backing or strengthening for receiving of subsequent finishes per finishing schedule.

B. Application

1. Curtain wall perimeter joint smoke & fire seal

Requirement in COP for Fire Safety in Building 2011: C10.2 & E12.1

▪ Firestop

The continuous gap between the curtain wall insulation and the concrete slab edge is to be filled with Hilti CFS-SP WB[FS09] – a 'Hong Kong Fire Services Department' approved mineral fiber insulation material with a 2-hour fire resistant period.

The Firestop system should comply with the latest local regulations, included but not limited to Code of Practice for Fire Safety in Buildings 2011, Clause E12.1.

External facades should be tested in accordance with the following applicable standards: BS EN 1365-3:2006 or BS EN 1364-4:2007.

▪ Smoke seal

A 3mm thick smoke seal joint spray Hilti CFS-SP WB[FS09] needs to be provided on the mineral fiber insulation material as a continuous smoke seal between the slab edge and the curtain wall at each slab level and to vertical walls or column where there is a core wall between two separate compartments.

The smoke seal system should comply with the latest local regulations, included but not limited to Code of Practice for Fire Safety in Buildings 2011, Clause C10.2. D-stability duration and maximum leakage is not more than 25m³/h/m² at 25Pa at ambient temperature when tested in accordance with BS EN 12101-1.

2. Linear joint firestop (door frame, wall joint, top of wall joint, movement joint,)

Requirement in COP for Fire Safety in Building 2011: C11.2

The continuous gap occurs between door frame and wall, in wall joints, top of wall joints and movement is to be filled with Hilti CP606[FS02] / CP601S[FS01] – a 'Hong Kong Fire Services Department' approved fire rated sealant with a FRR at least of the intervening floor.

Gaps within/forms a fire rated wall or fire barrier should have no less than the F.R.R. of the wall/barrier itself.

The movement of the movement joint or precast façades requires a sealant that can cater to movement. The fire rated sealant depth needs to be taken notice of in order for the right performance:

	Acrylic based sealant			Silicon based sealant	
Joint width (mm)	0-5	16-20	21-30	0-15	16-100
Sealant depth (mm)	6	10	15	6	15
Movement Capability	10%			25%	

It is intended that (Hilti CP601S) **[FS01]** silicon based sealant should be used in outdoor area where acrylic based sealant (Hilti CP606) **[FS02]** should be used in indoor area.

3. Pipe penetration through fire barrier (plastic & insulated)

Requirement in COP for Fire Safety in Building 2011: C8.2

For combustible penetration service, including but not limited to insulated pipe/ plastic pipe, approved intumescent firestop material should be applied.

For insulated pipe with size of 10" or below that penetrate through fire barrier, approved intumescent fire wrap (Hilti CP648) **[FS03]** or intumescent fire collar (Hilti CP643 **[FS04]** [for pipes with sizes ranging from 1.5" – 6"]/ Hilti CP644 **[FS04]** [for pipes with sizes ranging from 7" to 10"]) should be applied.

For insulated pipe with size of 10" or above that penetrate through fire barrier, fire ring should be applied to terminate the insulated material.

For plastic pipe penetration through fire barrier approved intumescent fire wrap (Hilti CP648) **[FS03]** or intumescent fire collar (Hilti CP643/ Hilti CP644) **[FS04]** should be applied.

4. Pipe penetration through fire barrier (metal)

Requirement in COP for Fire Safety in Building 2011: C8.2

For non-combustible penetration service, including but not limited to metal pipe/ steel pipe/ copper pipe, approved firestop sealant should be applied (Hilti CP601S**[FS01]** / Hilti CP606**[FS02]**).

The fire rated sealant depth needs to be taken notice of in order for the right performance:

	Acrylic based sealant (Hilti CP606) [FS02]			Silicon based sealant (Hilti CP601S) [FS01]	
Joint width (mm)	0-15	16-20	21-30	0-15	16-100
Sealant depth (mm)	6	10	15	6	15

5. Air duct penetration

Requirement in COP for Fire Safety in Building 2011: C8.2

The opening between rectangular metal air duct and the edge of opening should be fully enclosed by fire safety coated mineral wool board (Hilti CP670) to provide F.R.R. no less than the fire rated wall/fire barrier itself.

Gaps between mineral wool board and concrete surface/metal sleeves to be fully filled by firestop sealant (Hilti CP601S**[FS01]** /CP606**[FS02]**)

6. Trunking penetration

Requirement in COP for Fire Safety in Building 2011: C8.2

Where cable trunking penetrates fire barrier, intumescent firestop material (Hilti CFS-BL**[FS07]**) should be installed/filled at the cross point of the trunking and wall/slab, to ensure while fire occurs the material will fully enclose the opening, considering the cable will also melt.

7. Socket box (concrete wall & drywall)
Requirement in COP for Fire Safety in Building 2011: C8.4
For openings in concrete wall for the installation of switch box/socket box/junction box, intumescent fire rated backings of linings (Hilti CP617[FS06]) must be provided to the back of the box to reinstate fire rating and/or rating of the concrete wall.
8. Lift frame, architrave, push button and indicator firestop
Requirement in COP for Fire Safety in Building 2011: C9.1
Penetrations in a liftwell including but not limited to indicator panels and lift call buttons should be protected by firestop material.
Lift door (including door frame) should have an FRR. Firestop mortar (Hilti CP636[FS05]) should be provided to infill the back void of lift door frame.
Cable bundle
9. Requirement in COP for Fire Safety in Building 2011: C8.2
Where cable bundle penetrates fire barrier, intumescent firestop material (Hilti CP611A) should be installed/filled at the cross point of the cable and wall/slab, to ensure while fire occurs the material will fully enclose the opening, considering the cable will also melt.

3.04 CLEANING

- A. Clean all items and adjacent surfaces roof work. Examine all firestop work and repair all damage. In the event damage is irreparable, remove and replace such items at no additional cost. Leave entire Work in condition acceptable to the Architect.
- B. The Contractor shall at all times keep the Project area free of excessive waste, materials and rubbish. The Contractor shall remove all such waste, equipment, other property and tools from the Project site. Legally and safely handle, transport, and dispose of waste and debris.

3.05 PROTECTION

- A. The Contractor shall continually protect firestop from the weather, damage and traffic during and after installation work. Install and provide temporary protection from work of other trades during Project Construction.
- B. Ensure that intumescent firestop and seal materials are not prematurely activated by exposure to fire or flammable materials.

END OF SECTION